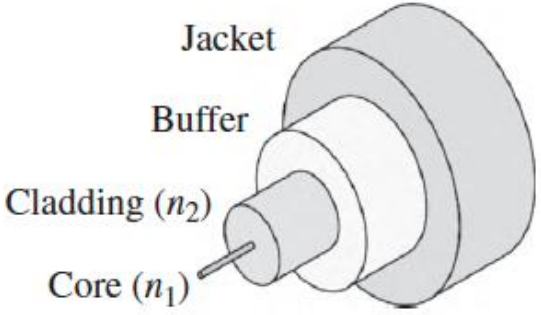
Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 16: Optical Fibers

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Dielectric waveguides

- Waveguides made entirely from dielectric materials are very desirable for applications where ohmic losses cannot be tolerated.
- The most familiar example is optical fiber, the dominant high-speed, high-capacity telecommunications transmission medium.



Optical fibers

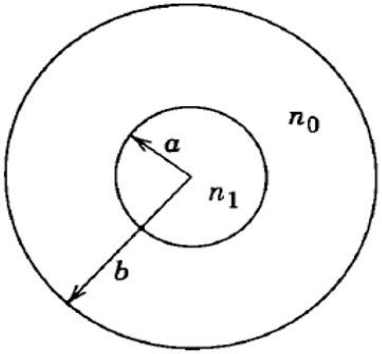
- Two classes:
  - **Multimode**
    - Cores are typically 50 μm in diameter.
    - The diameter is much larger than the wavelength of light (<1 μm)
    - Semi-geometrical optics.
  - Single mode
    - Cores are typically 5 μm in diameter, which is comparable with wavelength of light.

Multimode

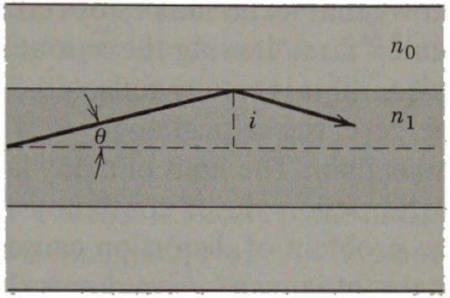
- Core: 2a = O (50 μm) with n1
- Cladding: 2b = O (150 μm) with n0 < n1
- The interface between the core and the cladding can be considered as flat.
- Critical angle for total internal reflection $i_0 = \arcsin(n0/n1)$
- $\theta < \theta_{max}$

$$\theta_{max} = \arccos\left(\frac{n_0}{n_1}\right) = \frac{\pi}{2} - i_0$$

Cross section



When i>i0



Waves Guided by Total Internal Reflection

- Define parameter Δ

$$\Delta = 1 - \frac{n_0}{n_1}$$ Difference between n_0 and n_1

A common value of Δ is 1%.

θ_{max} ? $\cos\theta_{max} = \frac{n_0}{n_1}$

$$1 - \frac{\theta_{max}^2}{2} = \frac{n_0}{n_1}$$

$$\theta_{max}^2 = 2\left(1 - \frac{n_0}{n_1}\right) = 2\Delta$$

$$\Rightarrow \theta_{max} = \sqrt{2\Delta}$$ if $\Delta = 1\%$, $\theta_{max} = 8$ deg

Wave Equation

- n(***x***) varies slowly in space of scale of the wavelength.

$$(\lambda \hat{k}) \cdot \nabla n << n$$

$$n = \sqrt{\frac{\varepsilon}{\varepsilon_0}} \qquad \varepsilon(\vec{x}) = \varepsilon_0 n^2(\vec{x})$$

Assume $e^{-i\omega t}$

No source

$$\Rightarrow \begin{cases} \nabla \cdot \vec{D} = 0 \\ \nabla \times \vec{E} = i\omega \vec{B} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{H} = -i\omega \vec{D} \end{cases} \qquad \begin{cases} \vec{B} = \mu_0 \vec{H} \\ \vec{D} = \varepsilon(\vec{x}) \vec{E} \end{cases}$$

Wave Equation

- n(***x***) varies slowly in space of scale of the wavelength.

$$\nabla \times \vec{E} = i\omega \vec{B} \quad \Rightarrow \quad \nabla \times (\nabla \times \vec{E}) = i\omega \nabla \times \vec{B}$$

$$\Rightarrow \quad \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = i\omega \mu_0 \nabla \times \vec{H} = i\omega \mu_0 \cdot (-i\omega \vec{D})$$

$$= \omega^2 \mu_0 \vec{D} = \omega^2 \mu_0 \varepsilon \vec{E}$$

$$\Rightarrow \quad \nabla^2 \vec{E} + \omega^2 \mu_0 \varepsilon \vec{E} - \nabla(\nabla \cdot \vec{E}) = 0$$

New wave equation.

Wave Equation

- n(***x***) varies slowly in space of scale of the wavelength.

Wave Equation

- n(***x***) varies slowly in space of scale of the wavelength.

$$\nabla \times \vec{H} = -i\omega \vec{D} \quad \Rightarrow \quad \nabla \times (\nabla \times \vec{H}) = \nabla \times (-i\omega \vec{D})$$

$$\nabla^2 \vec{H} - = -i\omega (\nabla \vec{D})$$

$$= -i\omega (\vec{D} \times \nabla \varepsilon + \varepsilon \nabla \times \vec{D})$$

$$= -i\omega (\vec{D} \times \nabla \varepsilon + i\omega \mu \varepsilon \vec{H})$$

$$\Rightarrow \quad \nabla^2 \vec{H} + \omega^2 \mu \varepsilon \vec{H} - i\omega \nabla \varepsilon \times \vec{D} = 0$$

Eikonal

Drop $\nabla\varepsilon$ in the wave eq.

$$\Rightarrow \quad \left[\nabla^2 + \frac{\omega^2}{c^2} n^2(\vec{x})\right]\psi = 0 \qquad \psi = E_i, H_i \quad i = x, y, z$$

$$\frac{n^2}{c^2} = \mu_0 \varepsilon(\vec{x})$$

Local wavenumber, $|\vec{k}(\vec{x})| = \frac{\omega}{c} n(\vec{x})$

Introduce $\psi = \psi_0 e^{i\omega S(\vec{x})/c}$

$S(\vec{x})$ is called eikonal. $S: [L]$

Eikonal approximation

$$\left[\nabla^2 + \frac{\omega^2}{c^2} n^2\right] e^{i\omega S/c} = 0$$

$$\nabla e^{i\omega S/c} = \frac{i\omega}{c} e^{i\omega S/c} \nabla S(\vec{x})$$

$$\Rightarrow \quad \nabla \cdot \left(\nabla e^{i\omega S/c}\right) = \nabla \cdot \left(\frac{i\omega}{c} e^{i\omega S/c} \nabla S(\vec{x})\right)$$

$$= \frac{i\omega}{c} e^{i\omega S/c} \nabla^2 S + \frac{i\omega}{c} \nabla S \cdot \frac{i\omega}{c} e^{i\omega S/c} \nabla S(\vec{x})$$

$$= \frac{i\omega}{c} e^{i\omega S/c} \nabla^2 S - \frac{\omega^2}{c^2} e^{i\omega S/c} (\nabla S) \cdot (\nabla S)$$

Eikonal approximation

$$\frac{iw}{c} e^{iwS/c} \nabla^2 S - \frac{w^2}{c^2} e^{iwS/c} (\nabla S)\cdot(\nabla S) + \frac{w^2}{c^2} n^2 e^{iwS/c} = 0$$

$$\Rightarrow \quad \frac{w^2}{c^2}\left[n^2 - (\nabla S)\cdot(\nabla S)\right] + \frac{iw}{c}\nabla^2 S = 0$$

Neglect $\frac{w}{c}\nabla^2 S$

$$\Rightarrow \quad n^2 = (\nabla S)\cdot(\nabla S) \qquad n(\vec{x}) = |\nabla S|$$

Eikonal Approximation

Eikonal approximation

$$S(\vec{x}) = S(\vec{x}_0) + (\vec{x} - \vec{x}_0) \cdot \nabla S(\vec{x}_0) + \dots$$

$$\psi = \psi_0 e^{\frac{i\omega S(\vec{x}_0)}{c}} \exp\left[i (\vec{x} - \vec{x}_0) \cdot \frac{\omega \nabla S(\vec{x}_0)}{c} \right]$$

$$= \psi_0 e^{\frac{i\omega S(\vec{x}_0)}{c}} \exp\left[i \vec{k}(\vec{x}) \cdot (\vec{x} - \vec{x}_0) \right]$$

$$\vec{k}(\vec{x}) = \omega \frac{\nabla S}{c}$$

∇S and $\vec{k}$ are in the same direction.

Eikonal approximation

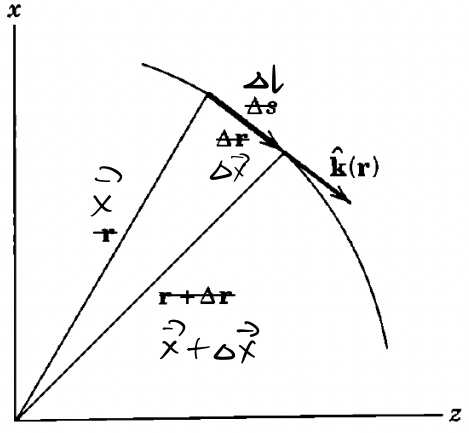
$$\Rightarrow \quad \nabla S = n(\vec{x})\,\hat{k}(\vec{x}) \qquad (1)$$

$$\lim_{\Delta l \to 0} \frac{\Delta \vec{x}}{\Delta l} = \hat{k}$$

l is the distance along the path.

$$\Rightarrow \quad \hat{k} = \frac{d\vec{x}}{dl}$$

$$(1) \Rightarrow \quad \nabla S = n \frac{d\vec{x}}{dl} \qquad (2)$$



Generalization of Snell's Law

$$\frac{d}{dl}\textcircled{2} \Rightarrow \quad \frac{d}{dl}\left(n\frac{d\vec{x}}{dl}\right) = \frac{d}{dl}\nabla S$$

$$\Rightarrow \quad \frac{d}{dl}\left(n\frac{d\vec{x}}{dl}\right) = \nabla\frac{dS}{dl} = \nabla\left(\frac{d\vec{x}}{dl}\cdot\frac{dS}{d\vec{x}}\right)$$

$$= \nabla(\vec{k}\cdot\nabla S) = \nabla n$$

$$\frac{d}{dl}\left(n\frac{d\vec{x}}{dl}\right) = \nabla n$$ Generalization of Snell's Law.

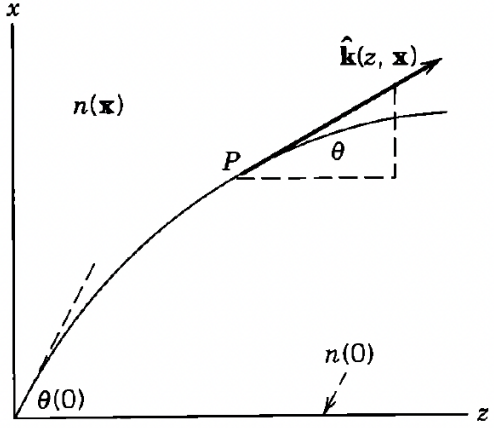
Transmission of meridional rays or equivalently to rays in a ""slab" geometry

$$\frac{d}{dL}\left(n\frac{d\vec{x}}{dL}\right) = \nabla n$$

$$\frac{d\vec{x}}{dL} = (\sin\theta,\ 0,\ \cos\theta)$$

$$\Rightarrow \begin{cases} \dfrac{d}{dL}(n\sin\theta) = \dfrac{dn}{dx} & ① \\ \dfrac{d}{dL}(n\cos\theta) = \dfrac{dn}{dz} = 0 & ② \end{cases}$$

Assume n only depends on x.



Transmission of meridional rays or equivalently to rays in a ""slab" geometry

$$\frac{d}{dl}(n\cos\theta) = 0$$

$\Rightarrow$ $n\cos\theta$ is a constant.

If $n \downarrow$, $\cos\theta \uparrow$

Define $\bar{n} = n(x_{max}) = n(0)\cos\theta(0)$

If $\theta(0) \uparrow$, $\bar{n} \downarrow$, $x_{max} \uparrow$,

$\bar{n}$ depends on $\theta(0)$.

